

Report of Alior & ING Bank Stock Closing Price Analysis with MS Excel and R

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Chapter 1

Introduction

The motivation for this project originates from my undergraduate studies, during which I attended a course in statistics. Although the course was taught by excellent instructors, I often encountered difficulties in understanding the practical purpose of the concepts being introduced. At that time, I was able to apply statistical methods formally, but I lacked a clear perspective on how these tools could be used in real-world contexts. As a result, many of the calculations I performed felt abstract and disconnected from practical applications.

It was only after completing the course that I began to realize the broader importance of statistics and its role in modern data-driven fields. This realization sparked my interest in pursuing a career in Data Science. By engaging in self-directed learning through online platforms such as DataCamp and educational materials from the *StatQuest* channel, I managed to connect theoretical knowledge into a more coherent framework. These additional resources not only reinforced the foundations I had acquired at university but also allowed me to explore the practical applications of statistical concepts in real-life scenarios.

Nevertheless, much of my exposure remained theoretical, and I felt the need to consolidate my skills by working on a complete, structured project. For this reason, I decided to revisit and expand upon one of the projects originally carried out during the statistics course. This time, however, the approach is significantly different: instead of focusing only on calculations, I aim to interpret the results in depth, critically evaluate the applied methods, and relate them to broader statistical reasoning. In this way, the project serves as a bridge between my academic background, online learning, and the practical demands of data analysis.

The central focus of the project is the statistical examination of stock market data from two major financial institutions: Alior Bank and ING Bank Slazki. More specifically, the project aims to analyze the behavior of their stock closing prices and corresponding returns, both from a univariate and bivariate perspective. The analysis addresses several key challenges. First, it investigates which continuous distribution models best describe the stock price data, allowing for meaningful interpretation of statistical characteristics such as skewness, kurtosis, and tail behavior. Second, it explores the relationships between the returns of the two institutions through covariance and correlation analysis, highlighting potential dependencies in their performance. Furthermore, the project applies linear regression to model the joint dynamics of the returns, followed by a critical assessment of the model's validity and explanatory power. Together, these elements create a comprehensive statistical framework for studying stock market behavior.

The data used in the project consists of historical daily closing prices for Alior Bank and ING Bank Slazki. The dataset was obtained from the Stooq financial database, a reliable and widely used source of stock market data. The period under study spans from January 1, 2023, to December 31, 2024, providing a two-year horizon that captures both short-term fluctuations and longer-term market tendencies. This dataset forms the foundation for the analyses carried out, supporting distribution fitting, dependency exploration, and regression-based modeling.

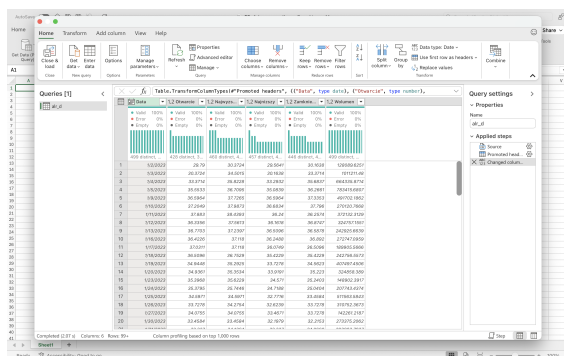
By combining theoretical knowledge with empirical analysis, the project ultimately aims to deliver both methodological insights and practical conclusions. It provides an opportunity to revisit core statistical concepts in a meaningful context, while at the same time contributing to my professional development as a future data scientist.

Chapter 2

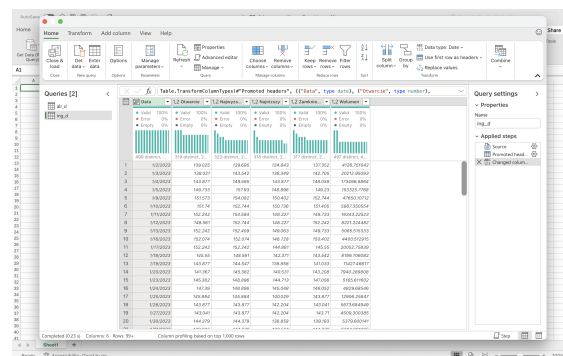
Data Preparation

2.1 Data Loading

Using **PowerQuery**, I imported the data from a CSV file. After analyzing the column descriptions, I confirmed that none of them contained errors or missing values. However, I quickly noticed that the dataset includes **missing days** — some dates are absent from the data. Interestingly, in both datasets (**alr_d**, **ing_d**), the missing days are exactly the same.



(a) Alior Bank company data

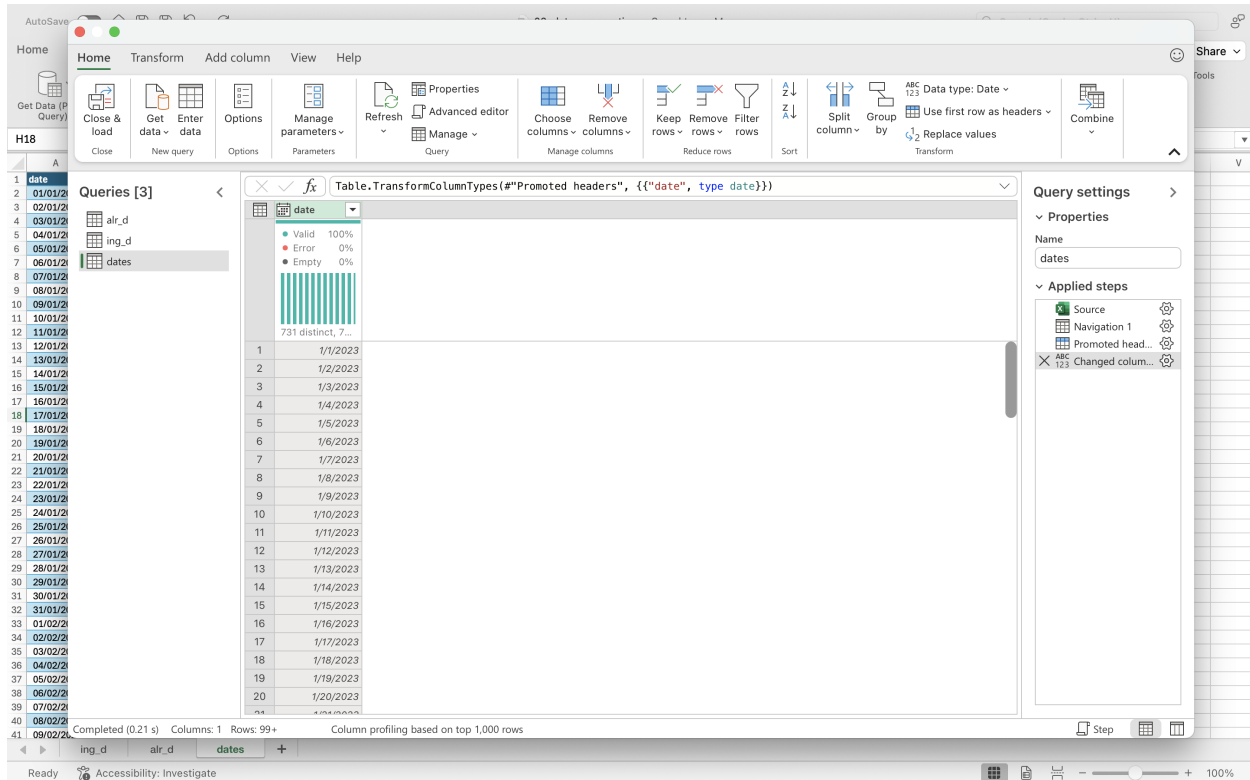


(b) ING Bank company data

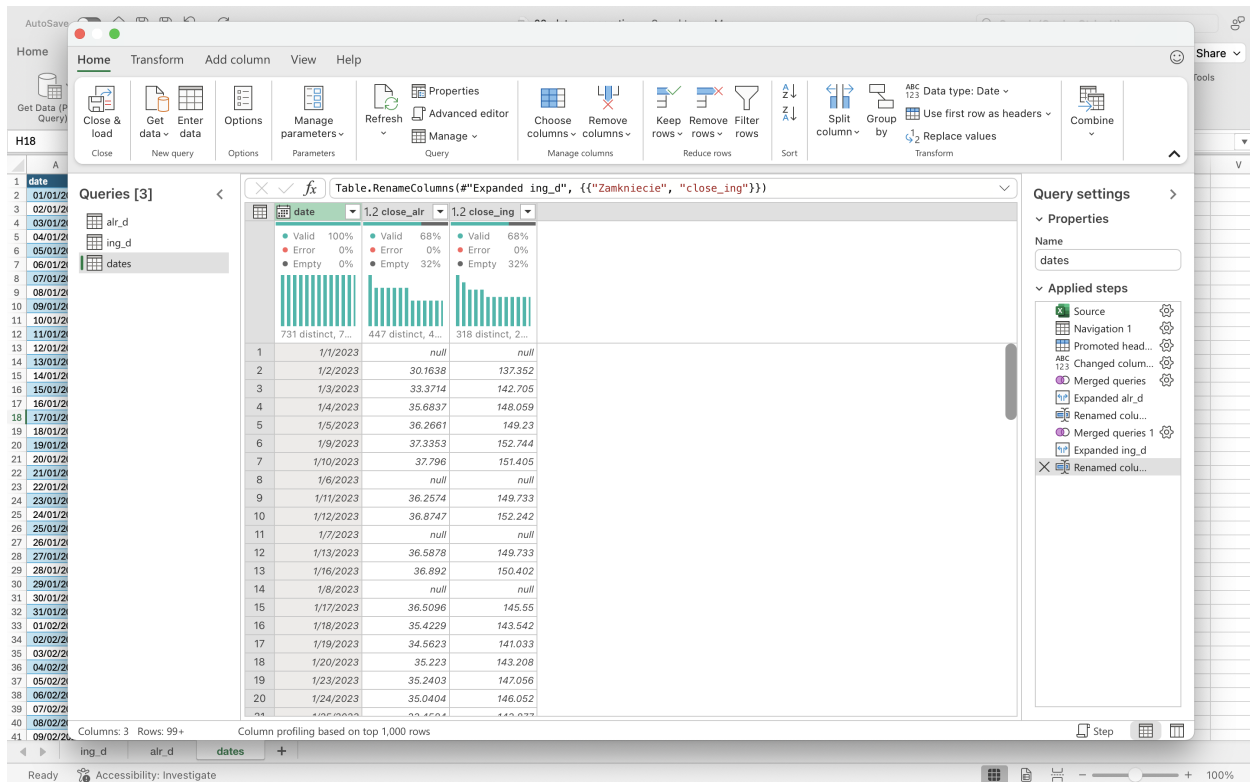
Figure 2.1: Initial closing price data for Alior and ING Bank.

2.2 Missing Data Analysis

I checked what percentage of the entire dataset is made up of missing days. If it were less than 5%, they could have been omitted. However, in our case, the missing data accounts for as much as 32%, so ignoring them could distort the entire distribution analysis.



(a) Generated set of all dates in the analyzed period



(b) Final dataset in Excel after importing all dates

Figure 2.2: Visualization of missing days and prepared dataset.

2.3 Data Preparation

The table with the required data was saved to a CSV file. Then, in **VSCode**, I replaced commas , with dots ., because Excel in my region saves numbers with a comma as the decimal separator.

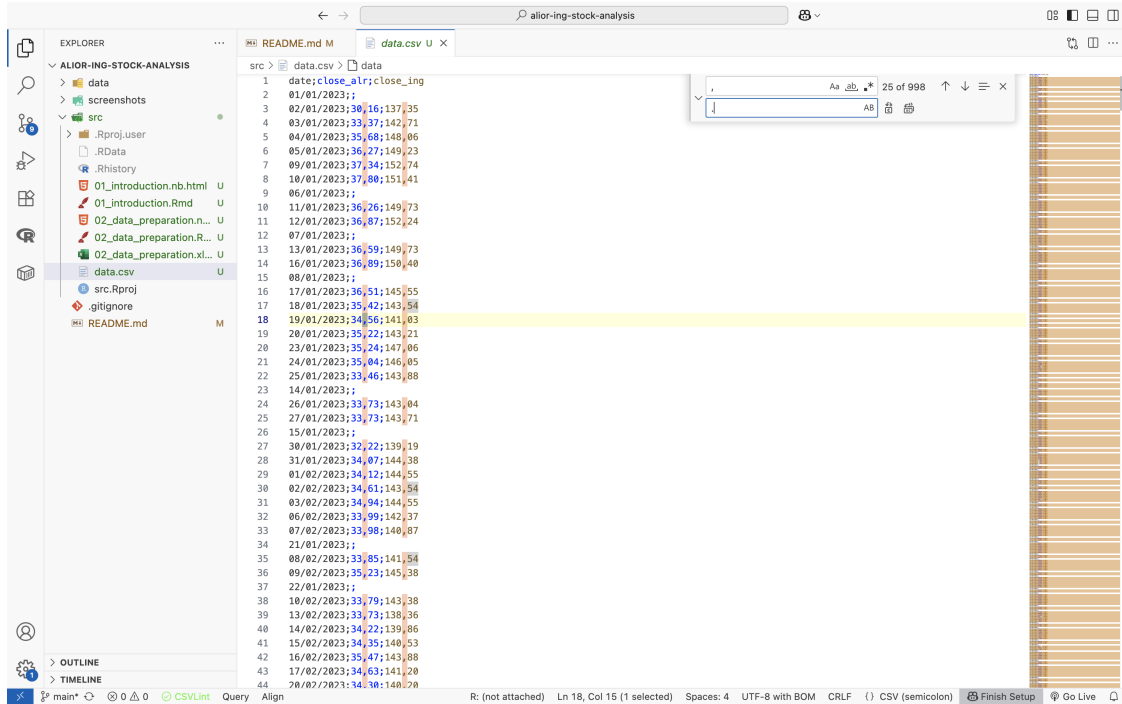


Figure 2.3: Replacement of decimal separator from comma to dot in VSCode.

2.4 Linear Interpolation of Missing Values

The next step is to perform **linear interpolation** of the missing data.

2.4.1 Definition of Linear Interpolation

Linear interpolation is a method to estimate missing values by assuming a linear change between two known points.

Mathematically, if a value y at position x is missing between two known points (x_0, y_0) and (x_1, y_1) , it is computed as:

$$y = y_0 + \frac{y_1 - y_0}{x_1 - x_0} \cdot (x - x_0)$$

Example Consider the dataset:

$$x = (3, 7, \text{NA}, 5, 8, \text{NA}, \text{NA}, 11)$$

1. The first missing value (**NA**) is between 7 (position 2) and 5 (position 4). Using the formula:

$$y = 7 + \frac{5 - 7}{4 - 2} \cdot (3 - 2) = 7 + (-1) = 6$$

2. The second missing value (**NA**) is between 8 (position 5) and 11 (position 8). Since there are **two consecutive NAs**, we interpolate them linearly:

- Position 6:

$$y = 8 + \frac{11 - 8}{8 - 5} \cdot (6 - 5) = 8 + 1 = 9$$

- Position 7:

$$y = 8 + \frac{11 - 8}{8 - 5} \cdot (7 - 5) = 8 + 2 = 10$$

After linear interpolation, the completed dataset becomes:

$$x = (3, 7, 6, 5, 8, 9, 10, 11)$$

2.4.2 Using Built-In Functions

In R, the `approx` function is used for linear interpolation.

The `approx` function works as follows:

- It takes a vector of known points:
 - `x`: representing indexes of non-empty values
 - `y`: representing values at the indexes in `x`
- It computes values at target points (`xout`) using **linear interpolation**, i.e., determining a point on the line connecting two neighboring known points.
- The parameter `rule = 2` ensures that values outside the range of known points are **extrapolated** instead of returning `NA`.

2.4.3 Saving the Interpolated Data

The interpolated data are saved into a CSV file to enable further analysis.

2.5 Summary

Thanks to this step, the dataset is now complete, with missing values filled using linear interpolation. This ensures that the data are ready for further analysis of the closing price distributions.

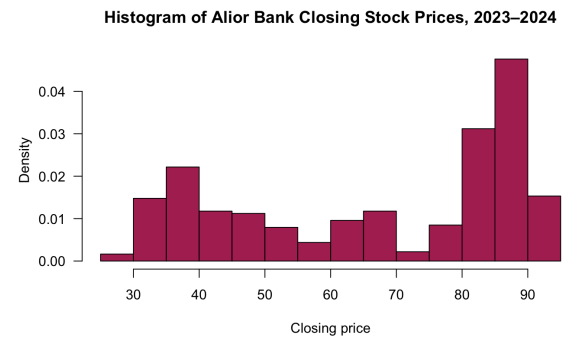
Chapter 3

Exploratory Data Analysis

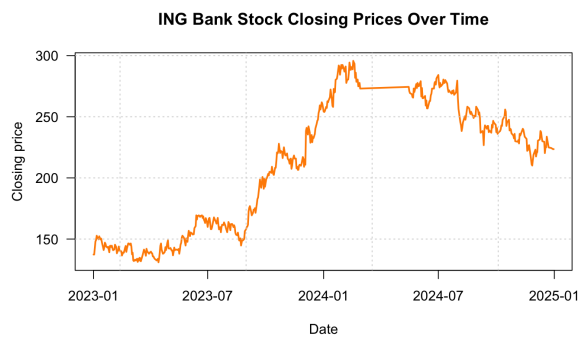
3.1 Time Series Visualization and Histogram



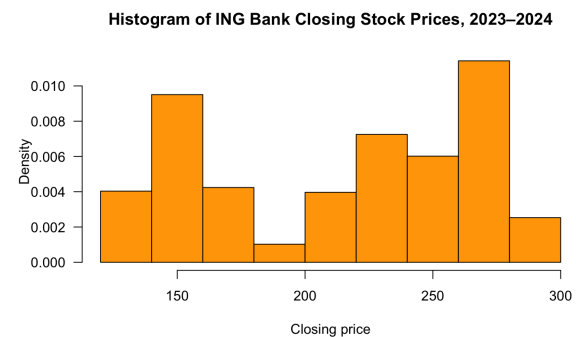
(a) Alior Bank Stock Closing Prices Over Time



(b) Histogram of Alior Bank Closing Prices, 2023–2024



(a) ING Bank Stock Closing Prices Over Time



(b) Histogram of ING Bank Closing Prices, 2023–2024

3.1.1 General Stock Price Trends

Analysis of price changes over time indicates a clear increase in Alior Bank's closing prices from the end of 2023 to the beginning of 2024. In 2023, prices rose gradually, while around the New Year there was a sharp jump. In 2024, Alior Bank prices remain relatively stable, fluctuating between 80 and 90, with drops quickly corrected by increases, indicating a maintained linear trend.

For ING Bank, closing prices in 2024 remain high, but unlike Alior Bank, they show a **clear downward trend** – from around 300 in January to below 250 in December 2024. Fluctuations in ING are also much larger, ranging from 210 to 290 throughout the year, although the amplitude of fluctuations gradually decreases.

3.1.2 Price Distribution Symmetry

The distribution of Alior Bank prices is left-skewed, meaning there is a long tail on the left side of the histogram. The highest number of observations is in the 80–90 range, corresponding to price stabilization in 2024.

ING Bank’s price distribution is also left-skewed, but in 2024 it shows an elongated distribution due to larger price fluctuations and the overall downward trend. Most observations are still in the 250–280 range, but in the second half of the year significantly lower values appear.

3.1.3 Fluctuations and Extreme Observations

For Alior Bank, a noticeable price increase is seen at the end of 2023 and the beginning of 2024. After this period, prices remain relatively stable, without extreme deviations during the year.

In ING, however, fluctuations in 2024 are much larger – prices vary from around 210 to 290. The scale of changes is significantly higher than for Alior Bank, and prices show a systematic decline over the year, which is clearly visible in the plot.

3.1.4 Preliminary Observations from the Visualizations

- Alior Bank prices rose gradually in 2023, with a sharp jump around the turn of 2023/2024. In 2024, prices remain in the 80–90 range.
- ING prices in 2024 remain high but show a clear downward trend – from around 300 in January to below 250 in December.
- Fluctuations in ING are much larger than in Alior Bank (210–290), while Alior Bank prices remain relatively stable.
- Both plots (time series and histograms) allow observing differences in price dynamics and volatility between the two companies.

3.2 Descriptive Statistics

Descriptive statistics for Alior Bank and ING Bank are presented in the following tables:

Statistic	Alior Bank
Mean	67.17
Median	69.0
Min	30.16
Max	90.0
Variance	290.12
SD	17.03
Skewness	-0.42
Kurtosis	2.55

Table 3.1: Descriptive statistics for Alior Bank (2023–2024)

Statistic	ING Bank
Mean	214.49
Median	220.0
Min	210
Max	300
Variance	1423.5
SD	37.73
Skewness	-0.35
Kurtosis	2.48

Table 3.2: Descriptive statistics for ING Bank (2023–2024)

3.2.1 Central Tendency

In 2023–2024, **ING Bank** exhibited **significantly higher closing prices** than Alior Bank. The average closing price for ING was 214.49 PLN, while for Alior it was 67.17 PLN. In both cases, the medians were noticeably higher than the means, suggesting a distribution with a longer left tail — i.e., a greater number of lower, extreme price values.

3.2.2 Dispersion

The range of stock prices indicates substantial differences. The highest price of Alior Bank shares was lower than the lowest price of ING shares, confirming that ING operates at a significantly higher price level. The standard deviation for ING is more than twice that of Alior, indicating a wider spread of data and higher price volatility.

3.2.3 Distribution Shape

Negative skewness values indicate that the price distribution has a longer tail on the left side, meaning there are more extreme drops than rises. This phenomenon is consistent with the observation that the median exceeds the mean. Both kurtosis values are below 3, indicating a platykurtic distribution — flatter than normal. In practice, this means that the stock prices of both banks are relatively stable, with fewer extreme deviations.

3.3 Monthly Returns Analysis

I conducted an analysis of the dynamics of ING Bank’s stock returns using a **waterfall chart** in Excel. This analysis allows us to see in which periods the stock value increased or decreased, and the contribution of individual months or quarters to the total return over a given period.

Stock returns are calculated using the standard formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\%$$

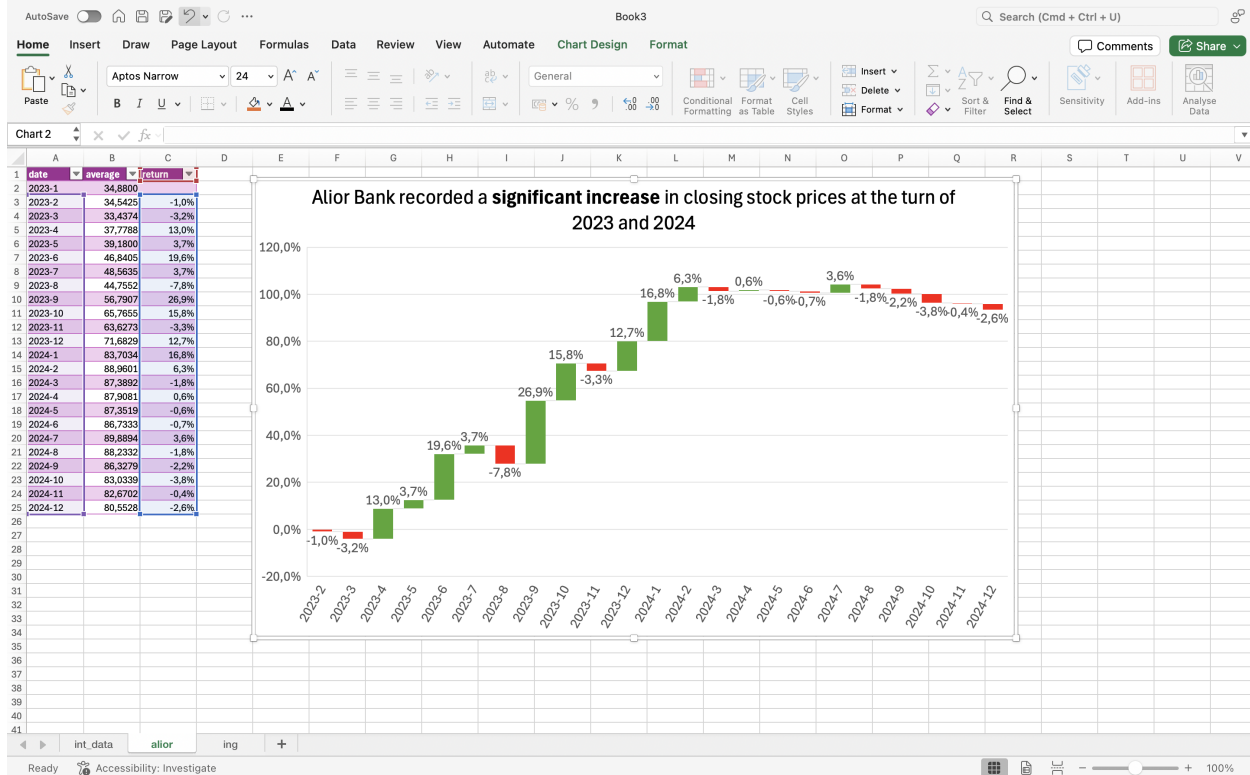
where: - R_t – return in period t , - P_t – closing price of the stock in period t , - P_{t-1} – closing price of the stock in the previous period.

Alternatively, if we want to express the return in ****currency (e.g., PLN)****, we can use the formula:

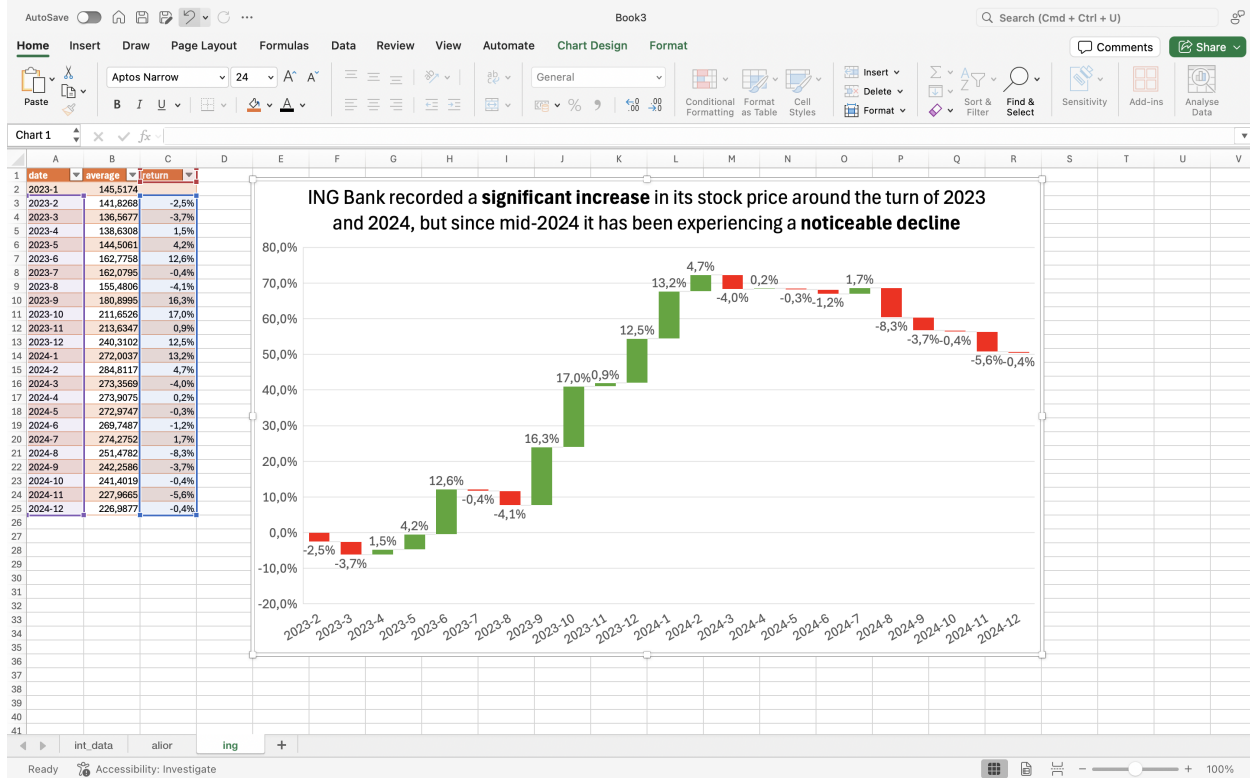
$$\Delta P_t = P_t - P_{t-1}$$

where ΔP_t is the direct change in the stock price during the period. These values can then be used in a waterfall chart to show which periods contributed most to the increase or decrease in stock value in PLN.

The chart in the file **waterfal_charts.xlsx** illustrates this dynamic, showing both positive and negative changes in stock prices over consecutive periods.



(a) Alior Bank Waterfall Chart



(b) ING Bank Waterfall Chart

Figure 3.3: Comparison of Waterfall Charts for Alior Bank and ING Bank.

3.3.1 Growth Dynamics

From September 2023 to February 2024, Alior Bank's stock price increased by 75%, while ING's rose by 64.6%. This indicates a very strong upward trend for both companies during this period.

3.3.2 Seasonality and Quarterly Dynamics

The waterfall chart analysis shows consistent growth in Q4 2023 and Q1 2024. For Alior Bank, prices remained stable in subsequent quarters, whereas ING experienced a noticeable decline during the same period. These observations confirm the earlier conclusions from the analysis.

3.3.3 Macroeconomic Context

The year 2023 was challenging for the economy, with GDP growth of only 0.2%. However, from Q4 2023 onwards, there was a rebound: in Q1 2024, the economy accelerated to 2.6–2.9% thanks to improved consumption and industrial activity. At the same time, inflation decreased from over 11% in 2023 to around 4% in 2024, reaching a minimum in March, which increased consumers' real purchasing power. Stable interest rates maintained by the Monetary Policy Council throughout 2024 provided predictable financial conditions and supported the growth of bank lending activity.

Chapter 4

Distribution Fitting

4.1 Fitting a Distribution to Data

One of the fundamental steps in data analysis is identifying a theoretical distribution that best describes the empirical data. This enables the subsequent application of statistical inference methods, tests, or predictive models that assume a specific distributional structure. In this chapter, several popular continuous distributions were fitted to data characterized by a right-skewed tail.

4.1.1 What is Distribution Fitting?

Distribution fitting is the process of finding a theoretical distribution (e.g., normal, log-normal, gamma, Weibull) that best describes empirical data. Goals:

- Estimating the distribution parameters based on the sample
- Assessing the quality of the fit
- Selecting the best model for further analysis and inference

4.1.2 Selected Distributions

Log-normal Distribution

If a random variable X follows a normal distribution:

$$X \sim N(\mu, \sigma^2),$$

then $Y = e^X$ follows a log-normal distribution. The probability density function (pdf) is:

$$f(y; \mu, \sigma) = \frac{1}{y \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right), \quad y > 0.$$

Parameters:

- μ – mean of the logarithms
- σ – standard deviation of the logarithms

Gamma Distribution

Probability density function:

$$f(x; k, \theta) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}, \quad x > 0,$$

Parameters:

- k – shape parameter
- θ – scale parameter

Weibull Distribution

Probability density function:

$$f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad x > 0,$$

Parameters:

- k – shape parameter
- λ – scale parameter

4.1.3 Parameter Estimation – Maximum Likelihood Estimation (MLE)

MLE selects distribution parameters that maximize the likelihood function:

$$L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta),$$

where θ is the vector of distribution parameters. Often, the log-likelihood function is used:

$$\ell(\theta) = \sum_{i=1}^n \ln f(x_i; \theta).$$

MLE finds such $\hat{\theta}$ for which $\ell(\theta)$ reaches its maximum.

4.1.4 The fitdistrplus Package

The **fitdistrplus** package in R provides a convenient framework for distribution fitting and analysis. Its main features include:

- Parameter estimation for a wide variety of distributions (e.g., using Maximum Likelihood Estimation, MLE),
- Visualization of the fitted distributions through histograms, QQ-plots, and cumulative distribution functions (CDFs),
- Goodness-of-fit testing to assess how well the theoretical distribution matches the empirical data,
- Model comparison using information criteria such as Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

4.1.5 Data Preparation

The analyzed data were transformed so that all values are positive. To achieve this, the entire dataset was shifted by adding the value $|\min(x)| + 0.001$, ensuring the absence of negative values and allowing the use of distributions defined only on the positive domain (e.g., Gamma, Weibull, Log-normal).

4.1.6 Methods

Distribution fitting was performed using **Maximum Likelihood Estimation (MLE)**. Parameters were estimated using the `fitdistr` function from the `fitdistrplus` package. Three distributions were considered:

- Log-normal,
- Gamma,
- Weibull.

For each distribution, the parameters were estimated and theoretical curves were fitted to the histogram of the empirical data. The quality of the fit was evaluated both visually and using statistical tests and information criteria.

4.1.7 Results

The histogram of the data was overlaid with the theoretical curves corresponding to each fitted distribution.

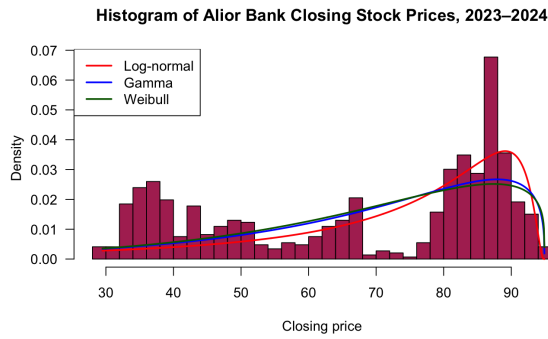


Figure 4.1: Estimated theoretical distributions for Alior Bank: Log-normal (red), Gamma (blue), Weibull (green)

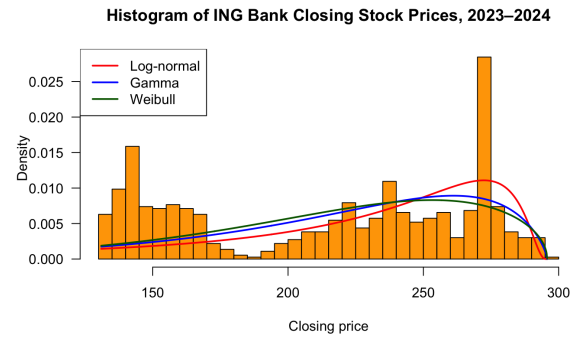


Figure 4.2: Estimated theoretical distributions for ING Bank: Log-normal (red), Gamma (blue), Weibull (green)

Preliminary visual analysis indicates that the Log-normal distribution is the best-fitting model. Its curve follows the empirical frequency points most closely, particularly in the central part of the distribution.

However, all three models show significant deviations in the left tail. The histogram indicates a higher concentration of observations in this region, which none of the tested models fully capture. This suggests that, while the overall fit is reasonable, the distributions may not completely represent the characteristics of the data, especially for extreme values.

4.2 Goodness-of-Fit Analysis

4.2.1 Interpretation of Goodness-of-Fit Statistics

To assess how well each theoretical distribution (Log-normal, Gamma, Weibull) fits the empirical data, the following test statistics were computed:

- **Kolmogorov–Smirnov (KS):** measures the maximum difference between the empirical and theoretical cumulative distribution functions (CDFs). The statistic is:

$$D_n = \sup_x |F_n(x) - F(x)|,$$

where $F_n(x)$ is the empirical cumulative distribution function (ECDF) and $F(x)$ is the theoretical CDF. Smaller values of D_n indicate a better fit.

- **Cramér–von Mises (CM)**: integrates the squared difference between empirical and theoretical CDFs. Lower values indicate a better fit. The statistic is:

$$W^2 = n \int_{-\infty}^{\infty} (F_n(x) - F(x))^2 dF(x).$$

- **Anderson–Darling (AD)**: a weighted version of CM that gives more weight to the tails of the distribution. The statistic is:

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n ((2i-1) [\ln F(X_{(i)}) + \ln(1 - F(X_{(n+1-i)}))]),$$

where $X_{(i)}$ are the ordered observations. This test places more emphasis on the fit in the tails compared to KS and CM.

- **Akaike’s Information Criterion (AIC)**: measures the trade-off between goodness of fit and model complexity:

$$AIC = 2k - 2 \ln(\hat{L}),$$

where k is the number of parameters and $\hat{L}$ is the maximum value of the likelihood function. Lower AIC indicates a better fit.

- **Bayesian Information Criterion (BIC)**: similar to AIC but penalizes model complexity more strongly, especially for larger sample sizes:

$$BIC = k \ln(n) - 2 \ln(\hat{L}),$$

where n is the sample size. Lower BIC indicates a better fit, with stronger penalization for models with more parameters.

Table 4.1: Goodness-of-fit statistics and criteria for Alior Bank

Statistic / Criterion	Log-normal	Gamma	Weibull
Kolmogorov–Smirnov	0.1226	0.1330	0.1316
Cramér–von Mises	2.7962	3.6375	3.9572
Anderson–Darling	18.746	22.029	23.559
Akaike Information Criterion (AIC)	6424.787	6289.029	6281.107
Bayesian Information Criterion (BIC)	6433.976	6298.218	6290.296

Table 4.2: Goodness-of-fit statistics and criteria for ING Bank

Statistic / Criterion	Log-normal	Gamma	Weibull
Kolmogorov–Smirnov	0.1453	0.1471	0.1494
Cramér–von Mises	2.8300	2.4323	2.6621
Anderson–Darling	20.022	17.829	19.336
Akaike Information Criterion (AIC)	7997.883	7779.604	7749.787
Bayesian Information Criterion (BIC)	8007.072	7788.792	7758.976

For both datasets, Alior Bank and ING, three theoretical distributions were fitted: **Log-normal**, **Gamma**, and **Weibull**. Parameter estimation was performed using **Maximum Likelihood Estimation (MLE)**, and the quality of fit was assessed with **goodness-of-fit tests** (Kolmogorov–Smirnov, Cramér–von Mises, Anderson–Darling) as well as **information criteria** (AIC, BIC).

For **Alior Bank**, the goodness-of-fit tests indicate that the **Log-normal distribution** provides the closest match to the empirical data, particularly in the central part of the distribution. KS measures the maximum deviation between the empirical and theoretical cumulative distribution functions, CM averages

the squared differences, and AD gives more weight to the tails. Despite this, none of the distributions fully capture the left tail, suggesting that extreme low values are underrepresented. When considering **AIC and BIC**, which penalize model complexity while rewarding likelihood, the **Weibull distribution** emerges as the most efficient overall model. This highlights that different criteria can lead to different conclusions: Log-normal fits the data better statistically, whereas Weibull offers a better balance between fit and simplicity.

For **ING**, the goodness-of-fit tests favor the **Gamma distribution**, which captures tail behavior more accurately. KS values are similar across all distributions, while CM and AD highlight Gamma as superior in describing the overall shape, including the tails. Again, none of the models perfectly represent the left tail. However, **AIC and BIC** select the **Weibull distribution** as the most balanced model, reflecting an optimal compromise between fit and parsimony.

Overall, no single distribution perfectly describes either dataset. Goodness-of-fit tests emphasize closeness to empirical distributions, especially in tails, while information criteria highlight model efficiency and complexity. Combining these perspectives, one can conclude that **Log-normal and Gamma distributions are better for statistical fit**, whereas **Weibull provides the most balanced and parsimonious model** across both datasets. Graphical inspection of histograms and fitted curves reinforces these conclusions, particularly regarding tail behavior.

4.3 Hypothesis Testing of Distribution Fit

The next step in the analysis is to formally test whether the fitted distributions provide an adequate representation of the empirical data. This is done by evaluating the **p-values** of the goodness-of-fit tests. A p-value quantifies the probability of observing a test statistic at least as extreme as the one computed from the sample, assuming the null hypothesis is true.

For a **two-sided test**, the p-value is defined as:

$$p\text{-value} = P(T \leq t_{\text{obs}}) + P(T \geq t_{\text{obs}})$$

where T is the random variable representing the test statistic under the null hypothesis, and t_{obs} is the observed value of the statistic.

In this context, the **null hypothesis** (H_0) states that the empirical data come from the fitted distribution. If the p-value is sufficiently high, we do not reject H_0 , which would suggest that the distribution could plausibly describe the data.

However, there are important caveats for using p-values in the context of **stock closing prices or log-returns**:

- Financial data often exhibit **heavy tails, skewness, and autocorrelation**, which standard goodness-of-fit tests may not fully account for.
- A non-rejection of H_0 (high p-value) does not guarantee that the model captures all important features of the data, particularly extreme events in the tails.
- Conversely, very large sample sizes, as is common in financial time series, may produce extremely small p-values even for distributions that visually appear reasonable.

Thus, while p-values provide a formal test of fit, they should be interpreted **together with graphical inspection and information criteria**, rather than in isolation.

Bank	Distribution	p-value $\geq 5\%$	p-value $\geq 20\%$
Alior Bank	Log-normal	TRUE	TRUE
Alior Bank	Gamma	TRUE	TRUE
Alior Bank	Weibull	TRUE	TRUE
ING	Log-normal	TRUE	TRUE
ING	Gamma	TRUE	TRUE
ING	Weibull	TRUE	TRUE

Table 4.3: KS-test results for fitted distributions

All goodness-of-fit tests performed for both **Alior Bank** and **ING** yielded **p-values below conventional thresholds** ($p < 0.05$ and $p < 0.2$). According to these tests, the theoretical distributions differ significantly from the empirical data in every case. In other words, **none of the fitted distributions can be formally accepted as perfectly describing the observed data.**

Despite this, some models appear more plausible than others when considering the relative performance across tests:

- For **Alior Bank**, the **Log-normal distribution** showed the closest alignment with the empirical data, both in central values and in the tails, compared to Gamma and Weibull.
- For **ING**, the **Gamma distribution** was the most reasonable among the three, particularly in capturing tail behavior.

Visual inspection of the histograms suggests that **both datasets might be better represented by a mixture of distributions**, reflecting the presence of extreme values and possible multiple regimes. However, modeling such mixed distributions would require a **much more extensive analysis**, including identification of component distributions and estimation of their parameters, which goes beyond the current scope.

Conclusion: While formal tests reject all simple distributional models, Log-normal (Alior) and Gamma (ING) provide the most reasonable approximations, serving as a practical baseline for further analysis, albeit with clear limitations in the tails and extreme values.

Chapter 5

Linear Regression

5.1 Returns and Log>Returns

In financial analysis, **returns** and **log-returns** play a key role in measuring the relative changes in asset prices over time.

5.1.1 Simple Returns

A **simple return** calculates the relative change in price between consecutive periods:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where P_t is the price at time t , and P_{t-1} is the price in the previous period. Simple returns are intuitive, but they **do not add up linearly over time**, which can complicate multi-period analysis.

5.1.2 Log>Returns

A **log-return** is defined as:

$$r_t = \ln \frac{P_t}{P_{t-1}} = \ln(P_t) - \ln(P_{t-1})$$

The advantage of log-returns is their **additivity over time** – the log-return over multiple periods is the sum of the individual period log-returns. They are also often more suitable for modeling financial data, especially when prices grow exponentially.

5.1.3 Context in the Analyzed Data

In the data from **09/2023–02/2024**, there was a clear upward trend in the stock prices of both companies, which often moved **in parallel**. Therefore, a natural step is to analyze **log-returns**, which better highlight relative price changes and facilitate further statistical modeling.

Returns and log-returns were calculated in **Excel** and then imported into **R**, where **histograms** and a **scatter plot** comparing the log-returns of both companies were prepared. These visualizations allow for assessing the data distribution and the degree of interdependence between the companies.

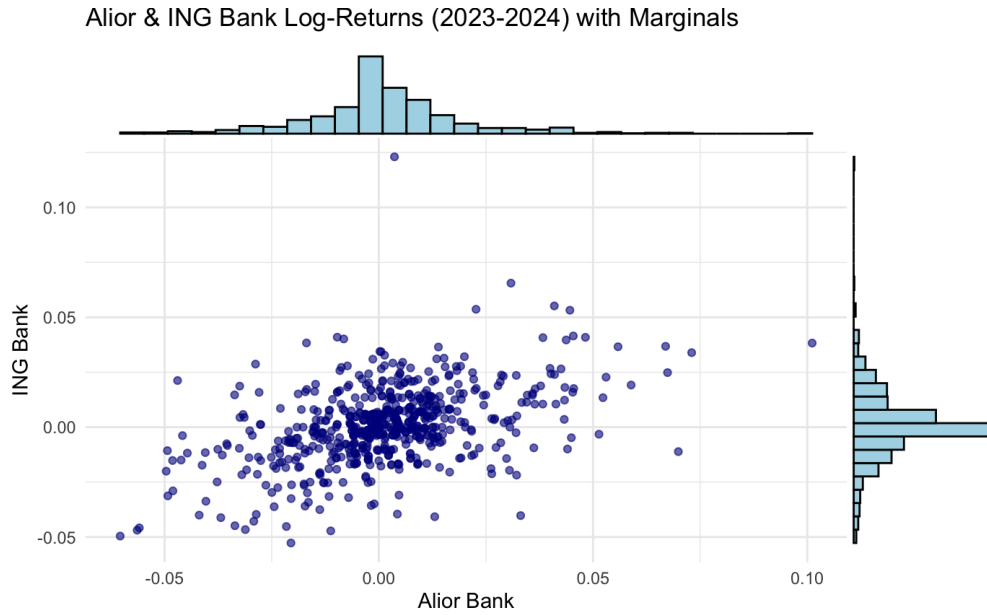


Figure 5.1: Scatter plot of Alior vs ING log-returns with fitted regression line.

5.2 Relationships Between Log-Returns

5.2.1 Covariance

Covariance measures how two variables change together:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- X_i, Y_i – observations of variables X and Y
- $\bar{X}, \bar{Y}$ – arithmetic means of the variables
- n – number of observations

A positive covariance indicates that the log-returns of Alior and ING tend to move together – when one stock rises, the other usually rises as well. A negative covariance would suggest that an increase in one stock corresponds to a decrease in the other. A value close to zero indicates little or no linear relationship. In the banking sector, positive covariance is common, as stock prices often respond similarly to macroeconomic factors and monetary policy.

5.2.2 Correlation

Correlation normalizes covariance, resulting in a value between $[-1, 1]$, which makes it easier to assess the strength of the relationship.

Pearson Correlation (P)

$$r_P = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Pearson correlation measures the linear relationship between Alior and ING log-returns. A value close to 1 means that changes in one stock's log-returns are strongly associated with changes in the other in the

same direction. A value close to -1 indicates a strong inverse relationship, and a value around 0 indicates little or no linear correlation. In our case, $r_P \approx 0.46$ shows a moderate positive linear relationship, meaning the stocks often move in the same direction but not perfectly in sync.

Spearman Correlation (S)

$$r_S = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Spearman correlation is based on the ranks of observations and measures monotonic relationships, without assuming linearity. A positive value indicates that higher log-returns of one stock generally correspond to higher log-returns of the other, even if the relationship is not strictly linear. In practice, $r_S \approx 0.43$ confirms a moderate common trend in Alior and ING log-returns, consistent with expectations for the banking sector.

5.2.3 Analysis of Log-Return Relationships Between Alior and ING

Analyzing covariance and correlation allows us to assess the extent to which the log-returns of two companies move together.

- **Positive covariance** indicates that Alior and ING log-returns tend to rise together. When one stock increases, the other generally also shows an upward movement.
- **Pearson correlation** ($r_P \approx 0.464$) shows a moderate positive linear relationship. The stocks' log-returns generally tend to move in the same direction, although some scatter exists and the relationship is not perfectly linear.
- **Spearman correlation** ($r_S \approx 0.429$) measures monotonic association and indicates that higher ING log-returns are usually associated with higher Alior log-returns, even if the relationship is not exactly linear.

In summary, both linear and monotonic analyses confirm a moderate positive dependence between Alior and ING log-returns, consistent with expectations for companies in the same banking sector.

$$\text{Cov}(\text{Alior}, \text{ING}) \approx 0.000135, \quad r_P \approx 0.464, \quad r_S \approx 0.429$$

5.3 Linear Regression of Alior Log-Returns on ING

The purpose of linear regression is to create a model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where β_0 is the intercept, β_1 the slope, and ε the random error term. This model allows us to describe the trend of the dependent variable Y (Alior log-returns) as a function of the independent variable X (ING log-returns).

5.3.1 Coefficient of Determination R^2

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

where $\hat{Y}_i$ are predicted values and $\bar{Y}$ the mean of observed values. The coefficient of determination $R^2 \in [0, 1]$ measures the proportion of variance in Y explained by the model.

5.3.2 Results of the Model

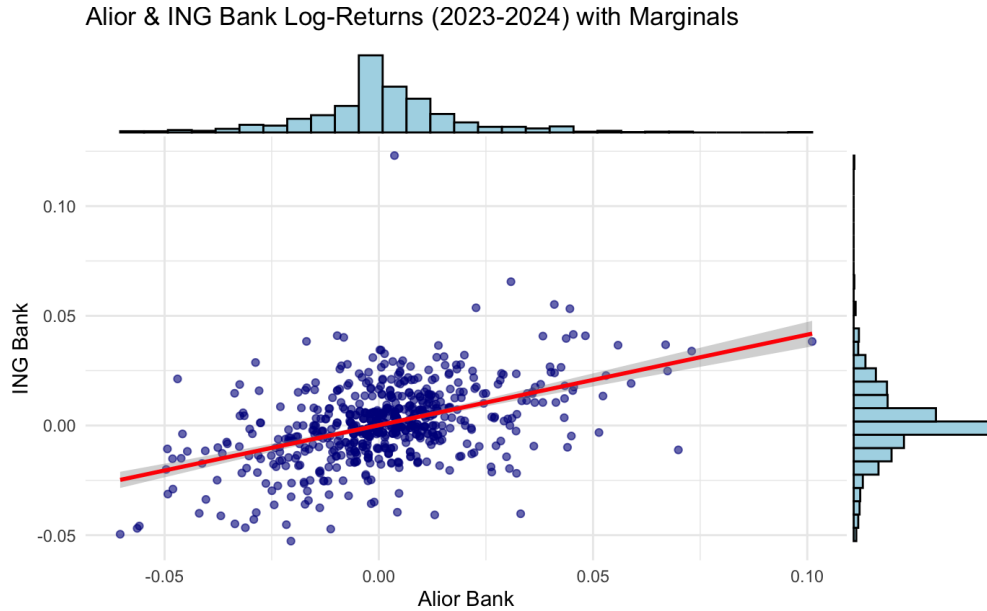


Figure 5.2: Residuals of the linear regression model for Alior log-returns on ING.

- **Intercept** $\hat{\beta}_0 \approx 0.00096$ – the average Alior log-return when ING log-return is zero. In practice, this means that if ING does not move, Alior remains practically unchanged.
- **Slope** $\hat{\beta}_1 \approx 0.5212$ – on average, a 1-unit increase in ING log-return is associated with a 0.52-unit increase in Alior log-return.
- **Coefficient of determination** $R^2 \approx 0.215$ – about 21.5% of the variability in Alior log-returns is explained by ING log-returns. This is a moderate fit, typical for financial data, where a large portion of variation is driven by random factors.

Interpretation

Although the regression confirms a statistically significant relationship between Alior and ING log-returns, the explanatory power is limited. Over 78% of the variance in Alior log-returns is due to other factors such as market noise, company-specific events, or macroeconomic conditions. This result is consistent with expectations for financial data, where returns are only partially predictable.

Chapter 6

Conclusion

The conducted analysis examined the dynamics of stock prices of Alior Bank and ING Bank in the years 2023–2024, with a focus on data quality, distribution fitting, and the interdependence between the two institutions. The study not only provided valuable insights into the financial data but also revealed a number of methodological challenges, highlighting the complexity of statistical modeling in real-world conditions.

The dataset contained a significant number of missing values in the closing prices of both banks. Ignoring these gaps could have substantially affected the results, while simple replacement with existing values would have distorted the distribution of changes. For this reason, I applied linear interpolation, which offered a more reliable way to reconstruct the missing information without introducing strong biases.

When examining the trends in stock prices, a clear upward movement was observed between the fourth quarter of 2023 and the end of the first quarter of 2024. From the second quarter of 2024 onward, the dynamics became more linear, which aligns with the macroeconomic conditions of that period. However, ING experienced a more pronounced decline than Alior Bank, suggesting differences in market response between the two institutions.

Further analysis focused on fitting theoretical distributions to the log-returns. The log-normal distribution showed the best fit for Alior Bank, while the gamma distribution performed best for ING. Nevertheless, all statistical tests produced p-values below 5%, which required rejecting the null hypothesis of conformity. The main source of mismatch lay in the left tails of the distributions, where extreme values were not well captured by the theoretical models.

The final stage of the study assessed the interdependence between the two banks. Visual inspection indicated some degree of association, which was confirmed by a regression model with $R^2 \approx 21.5\%$. This suggests that although the log-returns of Alior Bank and ING Bank are related, the strength of the relationship is insufficient to conclude that changes in one bank's stock can reliably explain changes in the other. It is therefore likely that additional external factors, not considered in this analysis, play a significant role.

Overall, the project allowed me to apply theoretical knowledge from the field of statistics to a practical financial context. It demonstrated that data modeling in real-world settings is far more complex than in classroom examples, requiring careful handling of missing values, critical assessment of model fit, and awareness of external influences. At the same time, the analysis strengthened my ability to use both Excel and R in practice, consolidating my skills in financial data analysis and providing valuable experience in bridging theory with application.